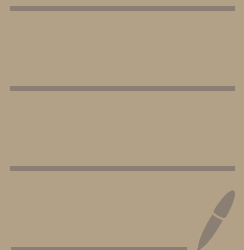


Ej Tena 20 Tipler



Dilación Térmica

26. ??

27. $\alpha = 1,7 \cdot 10^{-5}$ $L = 6 \text{ cm} = 0,06 \text{ m}$ $T_0 = 20^\circ \text{C}$
 $L_0 = 5,98 \text{ cm} = 0,0598 \text{ m}$

$$L = L_0 (1 + \alpha \Delta T)$$

$$\frac{6}{5,98} = 1 + 1,7 \cdot 10^{-5} (T_f - 20) \rightarrow T_f - 20 = 196,73$$

$$T_f = 216,73^\circ \text{C} = 489,88 \text{ K}$$

28. $\alpha_n = 1,2 \cdot 10^{-5} \text{ K}^{-1}$??

29. $T_0 = 20^\circ \text{C}$ $T_f = 60^\circ \text{C}$

$$V_0 = 1,4 \text{ L} = 1,4 \cdot 10^{-3} \text{ m}^3 \quad \Delta V = 7,5 \cdot 10^{-3} \text{ L} = 7,5 \cdot 10^{-6} \text{ m}^3$$

$$\Delta V = V_0 \beta \Delta T \rightarrow 7,5 \cdot 10^{-3} = 1,4 \cdot \beta \cdot (60 - 20)$$

$$\beta = 1,34 \cdot 10^{-4} \text{ K}^{-1} \quad \beta = 3\alpha \quad \alpha = 4,46 \cdot 10^{-5} \text{ K}^{-1}$$

30. $\alpha_{Ac} = 2,2 \cdot 10^{-5} K^{-1}$ $V_0 = 60L$ $T_0 = 10^\circ C$
 $\beta = 3\alpha \rightarrow \beta_{Ac} = 3,6 \cdot 10^{-5} K^{-1}$
 $\beta_{Hg} = 18 \cdot 10^{-5} K^{-1}$

$$V_{Ac} = V_0 (1 + \beta \Delta T) = 60 (1 + 3,6 \cdot 10^{-5} \cdot (25 - 10)) = 60,0324L$$

$$V_{Hg} = V_0 (1 + \beta \Delta T) = 60 (1 + 18 \cdot 10^{-5} \cdot (15)) = 60,162L$$

$$Derramado = 60,162 - 60,0324 = 0,1296L$$

32. $n = 1$ $T = 100^\circ C = 373,15K$ $P = 1 \text{ atm} = 1,013 \cdot 10^5 J$

$$a) PV = nRT \rightarrow V = \frac{8,314 \cdot 373,15}{1,013 \cdot 10^5} = 3,06 \cdot 10^{-2} m^3$$

$$b) \left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT$$

$b = 30 cm^3/mol = 30 \cdot 10^{-6} m^3/mol$
 $a = 9,55 \frac{J}{mol^2}$

$$\left(1,013 \cdot 10^5 + \frac{9,55 \cdot 1^2}{(3,06 \cdot 10^{-2})^2} \right) (3,06 \cdot 10^{-2} - 30 \cdot 10^{-6}) = 8,314 T \rightarrow T = 374,94K$$

34. $N = 1 \text{ átomo}$ $a = 0,03412 L^2/mol^2$ $b = 0,0237L/mol$
 $R = 0,08206 L \cdot atm / mol \cdot K$
 $N = n N_A \rightarrow n = 6,022 \cdot 10^{-23}$

$$\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT$$

$$\left(1 + \frac{0,03412 \cdot (6,022 \cdot 10^{-23})^2}{V^2} \right) (V - 6,022 \cdot 10^{-23} \cdot 0,0237) =$$